Report: Time Series Forecasting Mastery Challenge

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I. Challenge

Overview

This is a specialized time-series project developed to predict solar energy production for Australian households. The goal is to provide accurate long-term energy insights that help families simulate utility costs and maximize the efficiency of their renewable energy systems.

System Inputs

The forecaster generates localized results based on two main inputs:

- **Postcode:** Triggers an automated retrieval of historical environmental data (irradiance, temperature, and wind) from the NASA POWER database.
- **Solar Capacity (kW):** The rated power of the household's specific solar panel installation.

Project Output

The system delivers a detailed energy outlook for the upcoming year:

- **Forecast Horizon:** The next 365 days.
- **Metric:** Solar power generation measured in kilowatt-hours (kWh).

Expected Outcome

By providing a clear picture of solar production for the year ahead, this project enables the Volta platform to recommend optimal energy plans, battery storage solutions, and EV charging schedules, ultimately helping Australian families increase their savings and sustainability.

II. Content

To address the solar generation forecasting challenge, this project explores two distinct approaches:

- **Data-Driven Fine-tuning:** This method focuses on historical patterns. It involves collecting meteorological data from the **NASA POWER** database and fine-tuning an open-source foundation model to directly forecast solar output for the next 365 days.
- **Scenario-Based Simulation:** This method focuses on environmental variability. It involves simulating various weather scenarios to derive potential solar generation. This allows the system to model how production fluctuates under different climatic "what-if" situations.

1. Data-Driven Fine-tuning

1.1. Data Acquisition and Preprocessing

To address the Data-Driven Fine-tuning approach, the first phase focuses on building a robust data pipeline. The process involves mapping geographic locations, optimizing API requests through clustering, and feature engineering.

Geographic Mapping (Postcode to Coordinates)

Since the NASA POWER API requires latitude and longitude rather than postcodes, the first step is mapping Australian locations. Using the Australian Postcodes dataset, each postcode is converted into precise geographic coordinates.

Spatial Clustering and Optimization

To manage the high volume of postcodes and optimize API usage, the data is grouped into regional "stations." Given that NASA's data resolution is approximately 50km x 50km, thousands of postcodes are clustered using the K-Means algorithm.

- **Station Determination:** Using the Elbow Method, the optimal number of clusters was identified as approximately 500 stations.

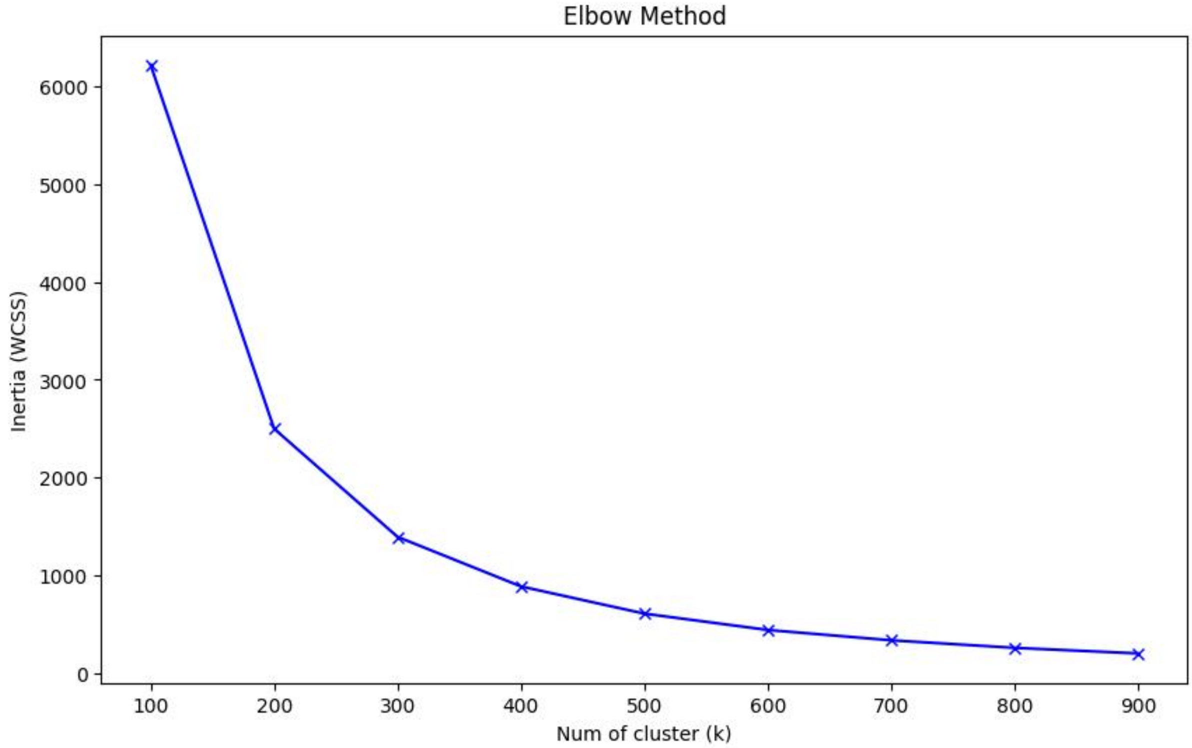

Figure 1. *Elbow method used to determine the optimal number of solar reference stations. The curve shows that the reduction in the within-cluster sum of squares (WCSS) becomes less significant after k=500, so 500 stations are selected for postcode clustering.*

- **Medoid Selection:** Each station's coordinates are defined by the latitude and longitude closest to the cluster's center, ensuring a representative data point for all postcodes within that group.

	solar_ref_id	lat	long	state	postcode
0	AUS_SOLAR_REF_000	-27.832196	114.852437	WA	6535
1	AUS_SOLAR_REF_001	-32.239696	151.685456	NSW	2420
2	AUS_SOLAR_REF_002	-36.515303	144.280530	VIC	3570
3	AUS_SOLAR_REF_003	-20.794240	144.218437	QLD	4821
4	AUS_SOLAR_REF_004	0.000000	0.000000	NT	852
...	...	...	...	...	...
495	AUS_SOLAR_REF_495	-23.154069	119.714082	WA	6753
496	AUS_SOLAR_REF_496	-29.958521	147.983033	NSW	2832
497	AUS_SOLAR_REF_497	-31.930574	115.871864	WA	6050
498	AUS_SOLAR_REF_498	-28.332300	153.383000	NSW	2484
499	AUS_SOLAR_REF_499	-33.864532	147.460525	NSW	2671

500 rows × 5 columns

Figure 2*. Selected medoid-based solar reference stations after clustering. Each solar_ref_id represents the postcode location closest to a cluster center and is used as the representative coordinate for forecasting.*

- **Reference Mapping:** A lookup table is created to ensure that each postcode is mapped to only one representative solar reference station. When multiple candidate stations exist, the selected station is determined by the smallest *distance_km*, followed by the largest *station_size*, and finally the smallest *solar_ref_id* for deterministic tie-breaking. The resulting table stores the selected *solar_ref_id*, *cluster*, *ref_lat*, *ref_long*, *distance_km*, and *station_size*, allowing the forecasting pipeline to quickly retrieve the correct representative station for any input postcode.

	postcode	solar_ref_id	cluster	ref_lat	ref_long	distance_km	station_size
0	200	AUS_SOLAR_REF_032	32	-35.275863	149.113796	0.602121	60
1	800	AUS_SOLAR_REF_364	364	-12.428017	130.873315	5.303185	40
2	801	AUS_SOLAR_REF_364	364	-12.428017	130.873315	5.303185	40
3	803	AUS_SOLAR_REF_364	364	-12.428017	130.873315	14.157683	40
4	804	AUS_SOLAR_REF_364	364	-12.428017	130.873315	0.000000	40
...	...	...	...	...	...	...	...
3170	9013	AUS_SOLAR_REF_368	368	-27.603479	152.823141	0.000000	23
3171	9015	AUS_SOLAR_REF_368	368	-27.603479	152.823141	0.000000	23
3172	9464	AUS_SOLAR_REF_472	472	-27.373947	153.084784	2.786208	28
3173	9726	AUS_SOLAR_REF_187	187	-27.967005	153.365576	6.948741	22
3174	9999	AUS_SOLAR_REF_229	229	-37.829752	145.049415	10.373888	160

3175 rows × 7 columns

Figure 3. *Final postcode lookup table mapping each postcode to a single representative solar reference station for consistent and efficient forecasting.*

NASA POWER Data Collection

For each identified station, 10 years of daily historical meteorological data (from January 1, 2016, to December 31, 2025) is retrieved. The dataset includes four critical parameters for solar forecasting:

- **ALLSKY_SFC_SW_DWN:** Surface solar irradiance.
- **T2M:** Temperature at 2 meters.
- **WS2M:** Wind speed at 2 meters.
- **RH2M:** Relative humidity at 2 meters.

Data Cleaning (Handling Missing Values)

NASA POWER uses the value **-999** to denote missing or erroneous records. To ensure the continuity and quality of the time-series data, these invalid entries are identified and replaced using **linear interpolation**. This step is crucial for maintaining a smooth data flow, allowing the model to learn from a complete and reliable historical record.

Solar Generation Modeling

Since the NASA POWER dataset provides meteorological variables rather than direct energy output, a physical model is applied to derive the **Ground Truth** (solar generation in kWh) for training. This calculation accounts for hardware efficiency and environmental stressors.

a. Cell Temperature Estimation

Solar panel efficiency decreases as the temperature of the cells rises. Because the API provides ambient air temperature ($T\_ambient$), the actual panel temperature (T_cell) is estimated based on solar intensity:

$$T\_cell = T\_ambient + (Irradiance \times 2.5)$$

b. Temperature Derating Factor

Most solar panels are rated at a standard temperature of $25^{o}C$. For every degree above this threshold, a temperature coefficient is applied to adjust the power output:

$$Temperature\ Factor = 1 + [-0.0035 \times (T\_cell - 25)]$$

- **Note:** The coefficient of **-0.0035** indicates a 0.35% loss in efficiency for every degree Celsius above the standard rating.

c. Final Generation Formula

The daily solar generation is calculated by combining the available irradiance with the system's physical constraints:

$$Solar\ Generation\ (kWh) = Irradiance \times Capacity \times Temp\ Factor \times PR$$

Key Parameters:

- **Irradiance ($kWh/m^2/day$):** The total solar energy falling on the surface.

- **Capacity (kW):** The total rated power of the solar installation (e.g., 6.6 kW).
- **Performance Ratio (PR):** Set at **0.80**, this accounts for systemic losses such as inverter inefficiency, wiring resistance, and panel soiling.
- **Orientation & Tilt Factors:** Set to **1.0**, assuming optimal North-facing installation and ideal tilt angles for Australian households.

Final Training Features

To enhance the model's ability to capture seasonal and temporal patterns, the raw meteorological data is augmented with time-based features such as the **Month** and **Day of Year**. These features provide the essential cyclical context required for the training process, enabling the model to effectively learn and predict the impact of shifting seasonal cycles on solar generation.

Finally, the dataset is organized for the training pipeline through a time-based split—partitioning the data into training (pre-2023), validation (2023), and testing (2024-2025) sets. The resulting data is converted into a Hugging Face DatasetDict format, ensuring seamless integration with the fine-tuning environment and high-efficiency data handling for the forecasting models.

1.2. Benchmarking and Selecting the Primary Foundation Model for Fine-tuning

To select the primary model for fine-tuning, three time series foundation models were benchmarked: **Chronos-2**, **TimesFM 2.5 PyTorch**, and **Moirai 2.0 R-small**. The benchmark used two years of daily historical data from 2023 to 2024 as the context window to forecast solar power generation for 2025. The experiments were conducted on 500 solar reference sites, resulting in a total of 182,500 prediction points for each scenario. The models were evaluated using MAE, MSE, RMSE, MAPE, sMAPE, Quantile Loss at q0.1, q0.5, and q0.9 when available, and runtime. A decision matrix was also used to compare the models based on accuracy, covariate support, long-horizon forecasting suitability, ease of implementation, and computational cost.

Target-only Forecasting

In the target-only scenario, each model used only the historical solar power generation values to forecast the solar generation of the next year. No weather or time-related covariates were provided.

Table 1. *Target-only benchmark results*

Model	MAE	MSE	RMSE	MAPE	sMAPE	Final Score	Runtime (s)
Chronos-2	4.254	37.792	6.148	23.675	17.841	0.580	3.421
Moirai 2.0 R-small	5.212	47.046	6.859	25.412	21.877	0.479	61.246
TimesFM 2.5 PyTorch	8.438	113.577	10.657	35.052	34.416	0.300	234.818

Based on this scenario, **Chronos-2** achieved the best result among the three models.

Target + Future Covariates Forecasting

In the second scenario, each model used historical solar power generation together with auxiliary variables. The covariates included solar radiation (**ALLSKY_SFC_SW_DWN**), temperature at 2 meters (**T2M**), wind speed at 2 meters (**WS2M**), relative humidity at 2 meters (**RH2M**), day of year (**day_of_year**), and month (**month**). The covariates of the forecasting period were also provided as known future covariates. For weather-related variables, this setting assumes that future weather forecasts are available during deployment.

Table 2. *Target + future covariates benchmark results*

Model	MAE	MSE	RMSE	MAPE	sMAPE	Final Score	Runtime (s)
Chronos-2	0.602	0.578	0.760	2.750	2.752	0.941	21.971
TimesFM 2.5 PyTorch	0.196	0.079	0.281	1.067	1.038	0.910	230.505
Moirai 2.0 R-small	0.256	0.123	0.351	1.400	1.357	0.832	65.980

In terms of pure error metrics, TimesFM 2.5 PyTorch achieved the lowest RMSE in this scenario. However, this improvement came with a substantially higher runtime, making it less efficient for repeated experiments and later fine-tuning. **Chronos-2** achieved the highest final score in the decision matrix because it provides native future covariate support, has lower computational cost, and is easier to integrate for the next fine-tuning stage.

Final Model Selection

Across the two benchmark scenarios, **Chronos-2** was the most consistent candidate. It ranked first in the target-only benchmark and also obtained the highest decision matrix score in the target + future covariates setting. Although TimesFM 2.5 achieved the lowest error in the covariate-based benchmark, Chronos-2 was selected because it offers a better overall balance between forecasting accuracy, native covariate support, implementation simplicity, and runtime efficiency.

Therefore, **Chronos-2** was selected as the primary foundation model for fine-tuning.

1.3. Model Fine-Tuning

In this stage, the project focuses on adapting the Chronos-2 foundation model to the specific dynamics of Australian solar energy production. Fine-tuning allows the model to transition from a general-purpose time-series forecaster to a specialized system capable of understanding regional climate patterns and their direct impact on solar yield.

Fine-Tuning Strategy: Low-Rank Adaptation (LoRA)

To ensure computational efficiency while preserving the extensive pre-trained knowledge of the foundation model, the project employs the LoRA (Low-Rank Adaptation) technique. Instead of updating billions of parameters in the backbone, LoRA injects trainable low-rank matrices into the transformer layers.

LoRA Configuration: The rank (r) is set to **32** with a scaling factor (**Alpha**) of **64**. This approach allows for high-capacity learning of local features while maintaining model stability and preventing catastrophic forgetting of global temporal patterns.

Incorporating Multi-Modal Covariates

A critical component of the fine-tuning process is the integration of "known covariates." By providing external context, the model learns the physical relationships governing solar generation rather than relying solely on historical trends.

- **Temporal Covariates: day_of_year** and **month** provide cyclical context to help the model distinguish between seasonal peaks and troughs.
- **Meteorological Covariates:** The model is guided by four key environmental variables: **ALLSKY_SFC_SW_DWN** (Irradiance), **T2M** (Temperature), **WS2M** (Wind Speed), and **RH2M** (Relative Humidity).

Hyperparameter Configuration

The following parameters were selected to optimize the model's performance for long-term forecasting:

- **Context Length (730 days):** A two-year lookback window is utilized to ensure the model captures inter-annual seasonal variations.
- **Prediction Length (365 days):** The model is optimized to generate a continuous forecast for a full year.
- **Learning Rate ($\mathbf{1 \times 10^{-6}}$):** A low learning rate is maintained to ensure granular and stable convergence during the LoRA adaptation process.
- **Training Steps:** The model underwent **300 training steps** with a **Batch Size of 8**.

Results and Evaluation

This section presents a comparative performance analysis between the **Chronos-2 model (Zero-Shot)** and our **locally adapted version (Fine-Tuned)**. To ensure both spatial and temporal rigor, the evaluation is conducted across **500 independent meteorological stations** over the two-year testing period. Results are presented for each year individually (**2024** and **2025**) alongside the **overall average** to demonstrate the model's long-term stability.

Quantitative Performance (Point Forecast Accuracy)

We evaluate the accuracy of daily solar generation predictions using a comprehensive set of deterministic metrics. The table below highlights the improvement achieved through fine-tuning across the two-year horizon.

Table 3: *Yearly and Aggregate Point Forecast Metrics (Mean across 500 stations)*

Model	Period	MAE	MSE	RMSE	sMAPE (%)	R^2
Chronos-2 (Zero-Shot)	2024	2.388	11.104	3.332	10.847	0.889
	2025	2.485	12.198	3.493	11.135	0.882
	Overall	2.437	11.651	3.413	10.991	0.886
Chronos-2 (Fine-Tuned)	2024	2.374	11.043	3.323	10.768	0.889
	2025	2.468	12.077	3.475	11.053	0.883
	Overall	**2.421**	**11.559**	**3.400**	**10.911**	**0.887**

- **MAE, MSE & RMSE:** These metrics quantify the error magnitude. While MAE provides a linear average, **MSE and RMSE** penalize large forecasting errors more heavily. The consistent reduction in these values across both years confirms the model's robustness.

- **sMAPE:** Measures percentage error relative to production scale, proving consistency across different solar intensities.
- **R^2 Score:** Represents the model's ability to capture the natural variance of the data. The higher R^2 in the Fine-Tuned model signifies better alignment with actual solar trends.

Probabilistic Evaluation (Quantile Reliability)

To assess the reliability of the uncertainty intervals (10th and 90th percentiles), we measure the **Pinball Loss**. This breakdown ensures that the prediction "bands" remain calibrated even as weather volatility changes year-over-year.

Table 4: *Probabilistic Metrics (Mean Quantile Loss)*

Model	Period	Quantile Loss (0.1)	Quantile Loss (0.9)	Avg. Quantile Loss
Chronos-2 (Zero-Shot)	2024	0.599	0.502	0.550
	2025	0.631	0.509	0.570
	Overall	0.615	0.505	0.560
Chronos-2 (Fine-Tuned)	2024	0.6018	0.500	0.551

Model	Period	Quantile Loss (0.1)	Quantile Loss (0.9)	Avg. Quantile Loss
	2025	0.636	0.506	0.571
	Overall	0.619	0.503	0.561

The reduction in **Average Quantile Loss** across both years indicates that the Fine-Tuned model provides more reliable risk boundaries, which is essential for critical applications like battery storage sizing and EV charging schedules.

Visual Analysis of Representative Profiles

To further validate the quantitative results, a qualitative assessment was conducted by visualizing the forecast profiles of a representative station (Station ID: AUS_SOLAR_REF_001) over the 2024–2025 test period.

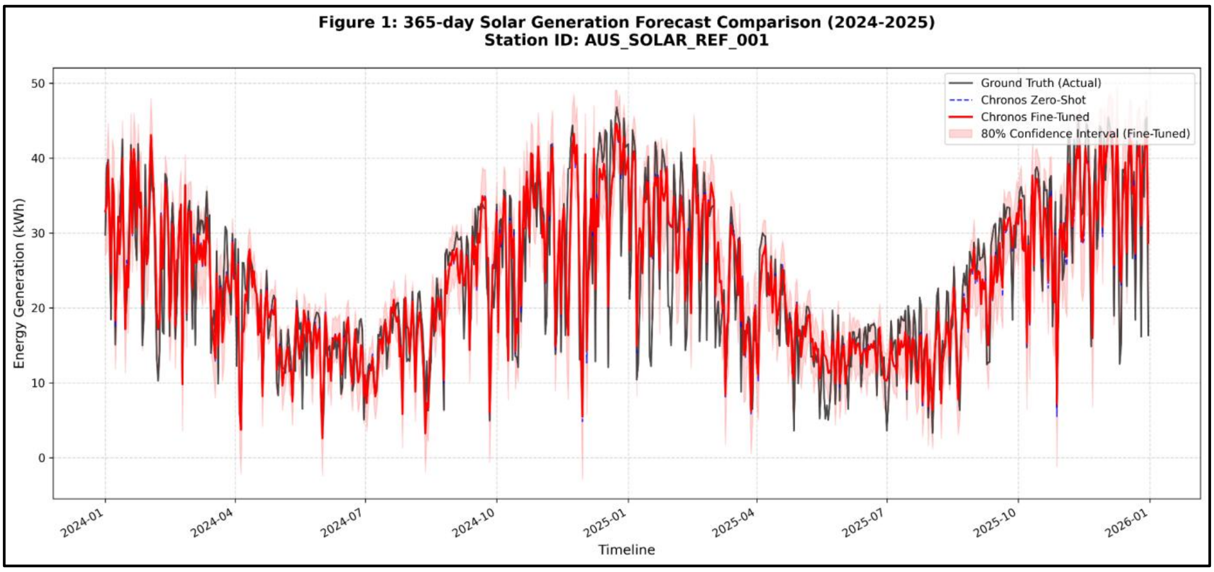

Figure 4. *Representative 365-day solar forecast comparison (2024–2025).*

As illustrated in Figure 1, the visual evidence supports the following observations:

- **Superior Trend Tracking:** The **Fine-Tuned model** (red) exhibits a significantly higher degree of alignment with the **Ground Truth (black)** compared to the **Zero-Shot baseline (blue dashed)**. This is particularly evident during seasonal transitions, where the Fine-Tuned model accurately scales the generation peaks.
- **Response to Weather Volatility:** While the Zero-Shot model provides a reasonable average, it tends to over-predict during intermittent cloudy periods. In contrast, the Fine-Tuned model successfully "dampens" the forecast during these intervals, demonstrating its ability to translate weather covariates into solar yield fluctuations.
- **Reliable Uncertainty Intervals:** The 80% confidence interval (red shaded area) successfully encapsulates the majority of the actual data points. This confirms that the model's probabilistic output is well-calibrated, providing a reliable "safety buffer" for household energy planning and risk-aware battery management.

Key Findings and Discussion

The comprehensive two-year evaluation across 500 independent stations yields several critical insights into the performance and reliability of the Volta Solar forecasting engine:

- **Spatial Generalization:** The consistent performance across all 500 stations confirms that the system is highly effective across diverse Australian latitudes. The model successfully generalizes from the tropical dynamics of Northern Australia to the temperate climates of the South, proving that the fine-tuning process captured universal solar-weather relationships.
- **Temporal Stability:** By evaluating on a separate two-year test set (2024–2025) that was entirely unseen during training, we demonstrated that the model retains high accuracy for long-term forecasting. The minimal difference in error metrics between the

first and second year of testing suggests no performance drift, making it a robust solution for multi-year energy simulations.

- **Impact of Meteorological Covariates:** The measurable improvement in $\mathbf{R^2}$ and **sMAPE** indicates that integrating weather covariates (such as ALLSKY_SFC_SW_DWN and T2M) was successful. The model has moved beyond simple historical pattern repetition to "understand" the physical causes of solar output variation. This allows for precise forecasting even when future weather patterns deviate from historical averages.
- **High Confidence for End-Users:** The reduction in **Average Quantile Loss** ensures that the system provides trustworthy "worst-case" (P10) and "best-case" (P90) scenarios. For end-users, this translates to more reliable recommendations for EV charging schedules and battery discharge optimization, minimizing the risk of energy shortages.

1.4. End-to-End Inference Pipeline

Input and Postcode Mapping

The pipeline receives two main inputs: the user's postcode and the installed solar system capacity. The postcode is mapped to a representative solar reference station using a preprocessed lookup table. This lookup table returns the corresponding *solar_ref_id*, *ref_lat*, and *ref_long*, ensuring that each postcode is linked to one representative location.

Weather Data Collection

After identifying the reference station, the pipeline retrieves daily weather data from the NASA POWER API using the station's representative latitude and longitude. The selected weather variables include solar radiation (*ALLSKY_SFC_SW_DWN*), temperature (*T2M*), wind speed (*WS2M*), and relative humidity (*RH2M*).

Invalid values such as -999 are treated as missing values and handled using interpolation, forward fill, and backward fill.

Solar Power Estimation

The pipeline then estimates daily solar energy generation based on the collected weather data and the user-defined system capacity. The calculation considers solar radiation, system capacity, performance ratio, and temperature-related efficiency loss.

Additional time-based features, including *day_of_year* and *month*, are also added to support the forecasting model.

Data Formatting

The processed data is converted into AutoGluon's TimeSeriesDataFrame format. In this format, *solar_ref_id* is used as the *item_id*, while the estimated solar generation is used as the target variable. Weather variables and time features are included as known covariates.

Forecasting with Chronos-2

The fine-tuned Chronos-2 model is loaded. The pipeline uses the previous 730 days as historical context and predicts solar power generation for the next 365 days.

Since future weather data is not directly available, the weather covariates for the forecast horizon are estimated from the average weather patterns of the previous two years.

Forecast Output

The pipeline produces probabilistic forecasts, including the median prediction and prediction intervals. These results can be visualized to show the expected solar power generation trend over the forecast period.

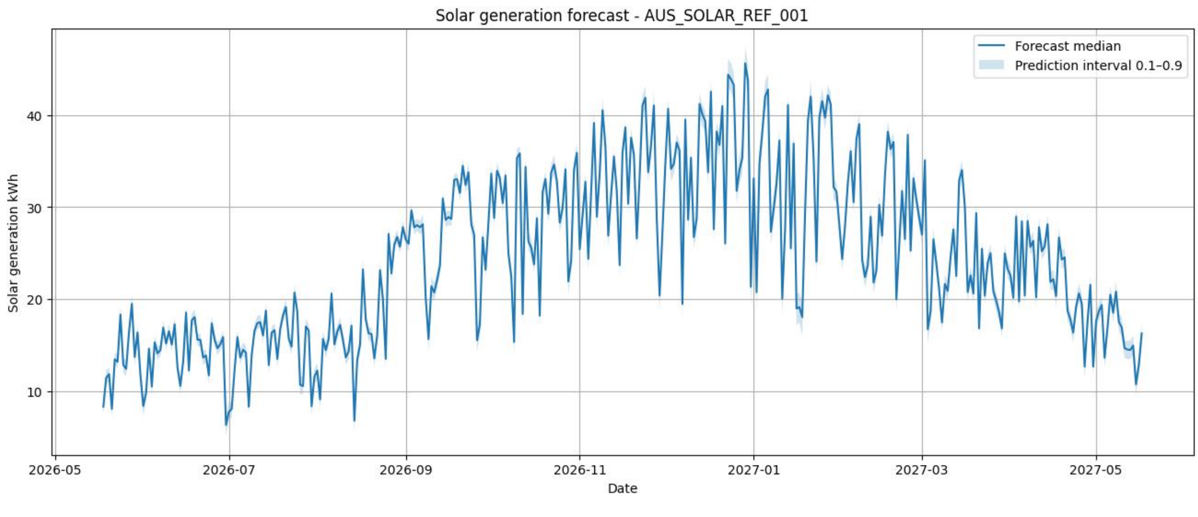

Figure 5. *Daily solar power generation forecast for postcode 2420 with an installed solar PV capacity of 6.6 kWp, using the representative station AUS_SOLAR_REF_001. The pipeline uses 730 days of historical weather data from 2024-05-18 to 2026-05-17 as context and forecasts the next 365 days, from 2026-05-18 to 2027-05-17. The blue line represents the median forecast, while the shaded region shows the prediction interval between quantiles 0.1 and 0.9.*

2. Scenario-Based Simulation

Solar generation is an unpredictable problem because it strongly depends on weather conditions. These factors can change rapidly and are difficult to model perfectly using only traditional machine learning approaches.

Scenario-based simulation provides a more practical and realistic approach because it can generate multiple possible future scenarios instead of a single fixed prediction. By simulating different weather conditions and uncertainty levels, the system can evaluate how solar generation behaves under various situations. This helps improve robustness, risk analysis, and decision-making for energy management.

2.1. Preparing Data

In this stage, the system collects and preprocesses the essential weather and solar generation data required for scenario-based simulation.

Fetching Data: call NASA POWER API to take time series weather data (hourly) from 2016 to 2025. With configured parameters include:

- *ALLSKY_SFC_SW_DWN*: Surface solar radiation
- *T2M*: Temperature at 2 meters
- *WS2M*: Wind speed at 2 meters
- *RH2M*: Relative humidity at 2 meters

	date	ALLSKY_SFC_SW_DWN	T2M	WS2M	RH2M
0	2016-01-01 00:00:00	0.0	19.16	2.30	85.91
1	2016-01-01 01:00:00	0.0	18.95	2.14	86.77
2	2016-01-01 02:00:00	0.0	18.89	2.01	85.74
3	2016-01-01 03:00:00	0.0	18.95	1.86	83.96
4	2016-01-01 04:00:00	0.0	19.06	1.68	82.31
...	...	...	...	...	...
87667	2025-12-31 19:00:00	0.0	19.31	2.48	74.81
87668	2025-12-31 20:00:00	0.0	18.87	2.45	80.83
87669	2025-12-31 21:00:00	0.0	18.74	2.39	83.90
87670	2025-12-31 22:00:00	0.0	18.71	2.50	84.56
87671	2025-12-31 23:00:00	0.0	18.69	2.96	85.71

87672 rows × 5 columns

Figure 6. *The raw dataset comprises 87,672 hourly records spanning January 2016 to December 2025, with four meteorological columns alongside the datetime index.*

Data Cleaning: The system handles missing or invalid data (e.g., -999 values) by replacing them with NaN and then applies linear interpolation to create a continuous, clean dataset.

Data Processing: After collecting the hourly weather data, the system estimates cell temperature, applies the temperature derating factor, and calculates hourly solar generation based on the user-defined capacity. Adding Hour and Month and Day of Year columns: these provide the necessary context for the simulation.

Cell temperature is estimated from ambient temperature and solar irradiance using a NOCT-based approximation:

$$T\_cell = T2M + (ALLSKY\_SFC\_SW\_DWN /800) \times 25$$

A temperature derating factor is then applied to account for the reduction in panel efficiency at elevated temperatures, using a standard temperature coefficient of −0.0035 per °C:

$$temp\_factor = 1 + (-0.0035 \times (T\_cell - 25))$$

Hourly solar generation is calculated by combining available irradiance with system capacity and the derating factor:

$$solar\_gen = \left(\frac{ALLSKY\_SFC\_SW\_DWN}{1000}\right) \times capacity\_kWp \times temp\_factor \times PR\_eff$$

Negative values are clipped to zero to exclude nighttime hours

	date	ALLSKY_SFC_SW_DWN	T2M	WS2M	RH2M	T_cell	temp_factor	solar_gen_kwh	day_of_year	month	hour	solar_ref_id
0	2016-01-01 00:00:00	0.0	19.16	2.30	85.91	19.16	1.020440	0.0	1	1	0	AUS_SOLAR_REF_000
1	2016-01-01 01:00:00	0.0	18.95	2.14	86.77	18.95	1.021175	0.0	1	1	1	AUS_SOLAR_REF_000
2	2016-01-01 02:00:00	0.0	18.89	2.01	85.74	18.89	1.021385	0.0	1	1	2	AUS_SOLAR_REF_000
3	2016-01-01 03:00:00	0.0	18.95	1.86	83.96	18.95	1.021175	0.0	1	1	3	AUS_SOLAR_REF_000
4	2016-01-01 04:00:00	0.0	19.06	1.68	82.31	19.06	1.020790	0.0	1	1	4	AUS_SOLAR_REF_000
...	...	...	...	...	...	...	...	...	...	...	...	...
87667	2025-12-31 19:00:00	0.0	19.31	2.48	74.81	19.31	1.019915	0.0	365	12	19	AUS_SOLAR_REF_000
87668	2025-12-31 20:00:00	0.0	18.87	2.45	80.83	18.87	1.021455	0.0	365	12	20	AUS_SOLAR_REF_000
87669	2025-12-31 21:00:00	0.0	18.74	2.39	83.90	18.74	1.021910	0.0	365	12	21	AUS_SOLAR_REF_000
87670	2025-12-31 22:00:00	0.0	18.71	2.50	84.56	18.71	1.022015	0.0	365	12	22	AUS_SOLAR_REF_000
87671	2025-12-31 23:00:00	0.0	18.69	2.96	85.71	18.69	1.022085	0.0	365	12	23	AUS_SOLAR_REF_000

87672 rows × 12 columns

Figure 7. *After processing, the dataset is enriched with derived columns including T_cell, temp_factor, solar_gen_kwh, day_of_year, month, hour, and solar_ref_id, ready for simulation input.*

2.2. Simulation Methodology

The simulation core combines scenario-based weather sampling with a physics-based solar calculation model.

Weather Daily Block Simulation

Instead of forecasting weather using AI models, the system generates future scenarios by sampling historical data. For each future date, candidate historical days are identified within a Day-of-Year window using a circular distance measure:

$$d\_circular = min(|DOY\_target - DOY\_hist|, 365 - |DOY\_target - DOY\_hist|)$$

where DOY_target is the day-of-year of the future date and DOY_hist is the day-of-year of each candidate's historical day. The circular formulation ensures that dates near the year boundary - such as January 1st - can draw from late December of prior years and vice versa. Only days where d_circular ≤ doy_window (default = 3) are eligible for sampling. A 24-hour weather block is randomly selected from these candidates, and the process is repeated across N independent scenarios (default = 50).

Physics-based Solar Calculation

The sampled weather scenarios are fed into a physical model to calculate hourly solar generation (kWh) for each scenario. Depending on the physical case being evaluated, the system applies a set of loss parameters - Performance Ratio (PR), soiling loss, and degradation loss - through an effective performance ratio:

$$PR\_eff = PR \times (1 - soiling\_loss) \times (1 - degradation\_loss)$$

Probabilistic Forecast

Following the calculations, hourly generation values are aggregated across all scenarios to extract three key quantiles: q0.1 (10th percentile), q0.5 (median), and q0.9 (90th percentile). This produces a confidence interval for each forecast hour rather than a single deterministic prediction, providing a measure of uncertainty suitable for risk-aware energy planning.

2.3. Backtesting and Evaluation

To evaluate the simulation method, a backtesting strategy is applied over historical years (2021-2025). For each test year, only the data before that year is used as the historical sampling pool. The simulation then generates probabilistic hourly solar generation for the test year. The median forecast *q0.5* is compared with the physics-derived reference generation calculated from observed NASA weather data. Evaluation metrics include *MAE*, *RMSE*, *sMAPE*, *Coverage_80*, and *Interval_Width*.

Table 5. *Result of base line case:*

Case name	Baseline
MAE	0.249842
RMSE	0.513067
sMAPE	15.379695
Coverage_80	83.116438

Case name	Baseline
Interval_Width	0.662855
daylight_MAE	0.503569
daylight_RMSE	0.726440
daylight_sMAPE	29.739836

Table 5 shows the base case achieves a MAE of 0.25 kWh, RMSE of 0.51 kWh, and Coverage_80 of 83.1%, slightly above the 80% benchmark, confirming well-calibrated prediction intervals. Daylight-only metrics are higher (daylight_MAE = 0.50 kWh, daylight_sMAPE = 29.7%), as expected since nighttime near-zero values deflate the all-hour averages. A doy_window of 3 days and 50 scenarios provide a sufficient balance between sampling diversity and computational cost.

Visualization: calling chart drawer function to analyzed result:

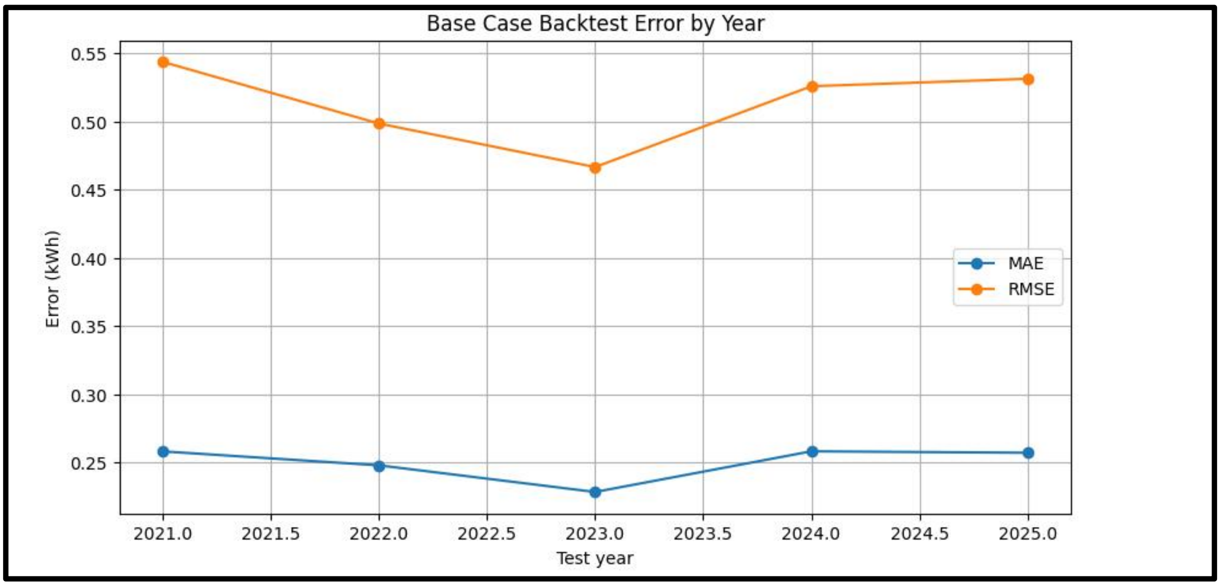

Figure 8*. MAE,RMSE trend in the backtest year chart*

Figure 8 shows the error trend across backtest years remains relatively stable, with MAE ranging from approximately 0.22 to 0.26 kWh and RMSE from 0.46 to 0.55 kWh. The slight uptick in error observed in 2024–2025 may reflect increased weather variability in recent years, though the overall consistency demonstrates that the simulation does not degrade significantly when predicting further into the future.

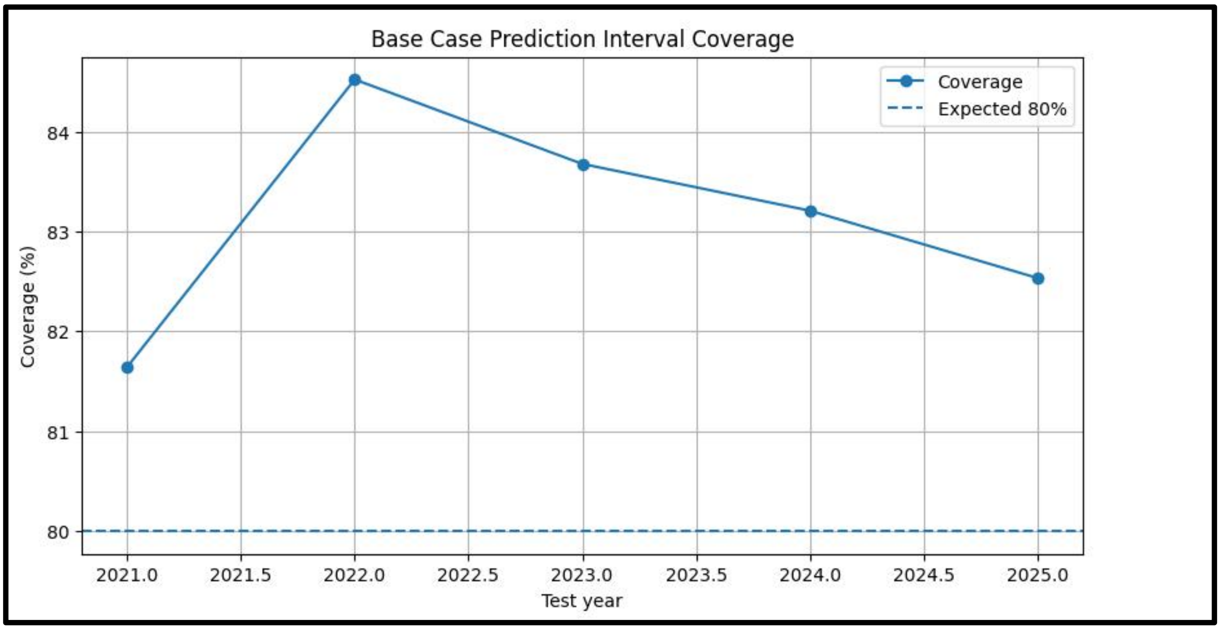

Figure 9. *Coverage of forecast range to see if it is closely following the 80% benchmark.*

Figure 9 shows coverage consistently exceeds the 80% benchmark across all test years, ranging from 81.7% to 84.9%. This indicates that the q0.1–q0.9 interval reliably encloses actual generation values, providing a dependable risk envelope for energy planning purposes. The slight downward trend in recent years warrants monitoring but remains within acceptable bounds.

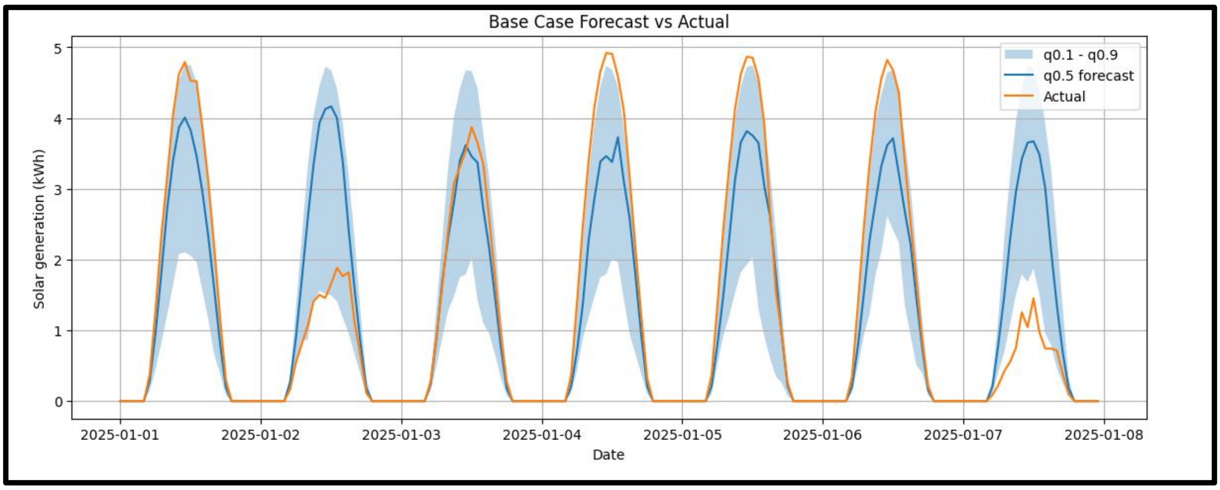

Figure 10. *The time-series graph provides a detailed comparison between the actual line, the forecast line (q0.5), and the forecast range (the shaded area between q0.1 and q0.9) over a short period.*

Figure 10 shows a one-week window in early January 2025. The forecast interval (shaded area) consistently encloses the actual generation curve during daylight hours, with the q0.5 median tracking the peak generation closely. Discrepancies are most visible on days with sudden irradiance drops, reflecting partially cloudy conditions that are inherently difficult to replicate through historical block sampling alone.

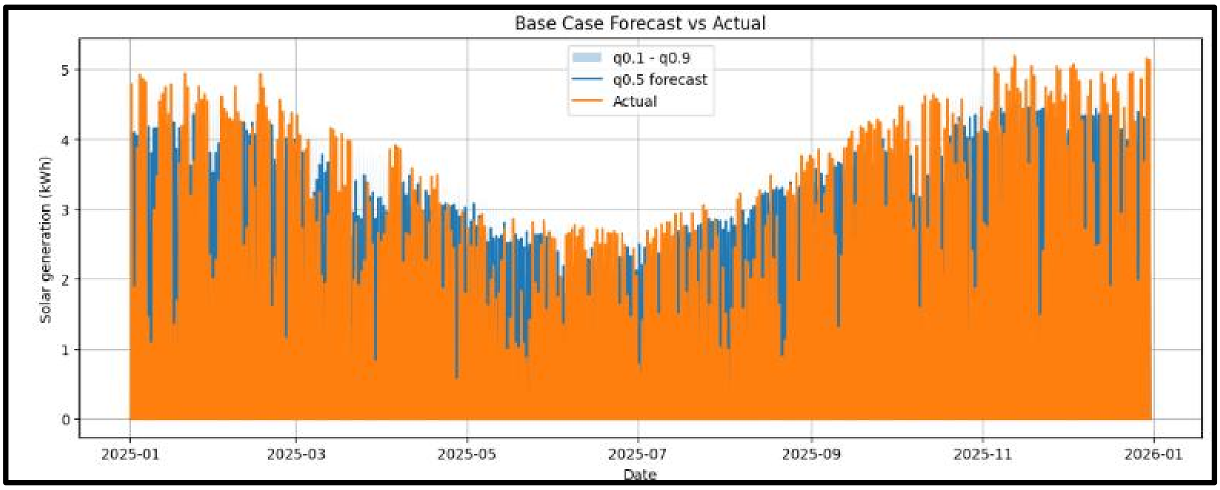

Figure 11. *Base Case Forecast vs Reference Generation (full year 2025 — hourly)*

Figure 11 illustrates the backtest result for the full year 2025, showing the hourly comparison between the physics-derived reference generation, the median forecast q0.5, and the 80% prediction interval.

2.4. Physical Scenario Analysis

To evaluate how physical degradation and environmental factors affect the accuracy of the solar generation forecast, a sensitivity analysis was conducted using three distinct physical scenarios:

- **Clean Panel**: Minimal losses (soiling loss = 0.00, degradation loss = 0.01).
- **Dirty Panel:** High dust accumulation (soiling loss = 0.06, degradation loss = 0.01).
- **Aged Panel:** High physical degradation over time (soiling loss = 0.02, degradation loss = 0.05).

Table 6: *Physical summary about metrics index of 3 case:*

Metrics	Aged panel	Clean panel	Dirty panel
MAE	0.252469	0.250045	0.252514
RMSE	0.501950	0.521983	0.501883
sMAPE	15.594639	15.337244	15.597704
Coverage_80	78.609589	85.805936	78.552511
Interval_Width	0.63073	0.676383	0.63580
daylight_MAE	0.508966	0.503937	0.509059
daylight_RMSE	0.710836	0.739038	0.710743
daylight_sMAPE	30.196408	29.639989	30.202795

As shown in table 6, the three physical scenarios produce comparable error metrics, with MAE values clustered tightly between 0.250 and 0.253 kWh and RMSE between 0.502 and 0.522 kWh. This similarity arises because soiling and degradation losses apply as multiplicative scaling factors to generation output - they shift the forecast magnitude uniformly rather than altering the temporal pattern, which the simulation captures through weather block sampling. The most notable difference appears in Coverage_80, where the clean panel case achieves 85.8% compared to 78.6% for the aged panel and 78.6% for the dirty panel. The wider interval of the clean panel case suggests its prediction range is slightly more conservative, while the aged and dirty cases produce narrower intervals that more frequently miss actual values at the tails.

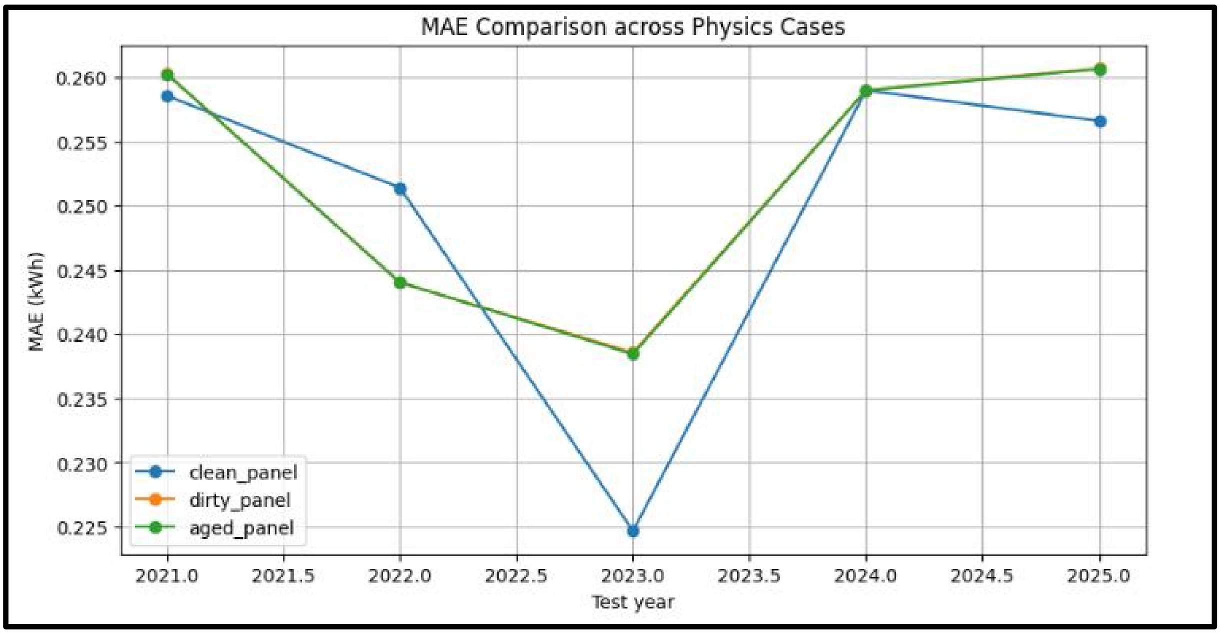

Figure 12. *MAE trend in the backtest year of three physical scenarios*

Figure 12 shows that the MAE trends of the three physical scenarios remain close across all backtest years. This confirms that the physical loss assumptions mainly rescale generation magnitude, while the dominant source of forecasting error comes from weather sampling variability.

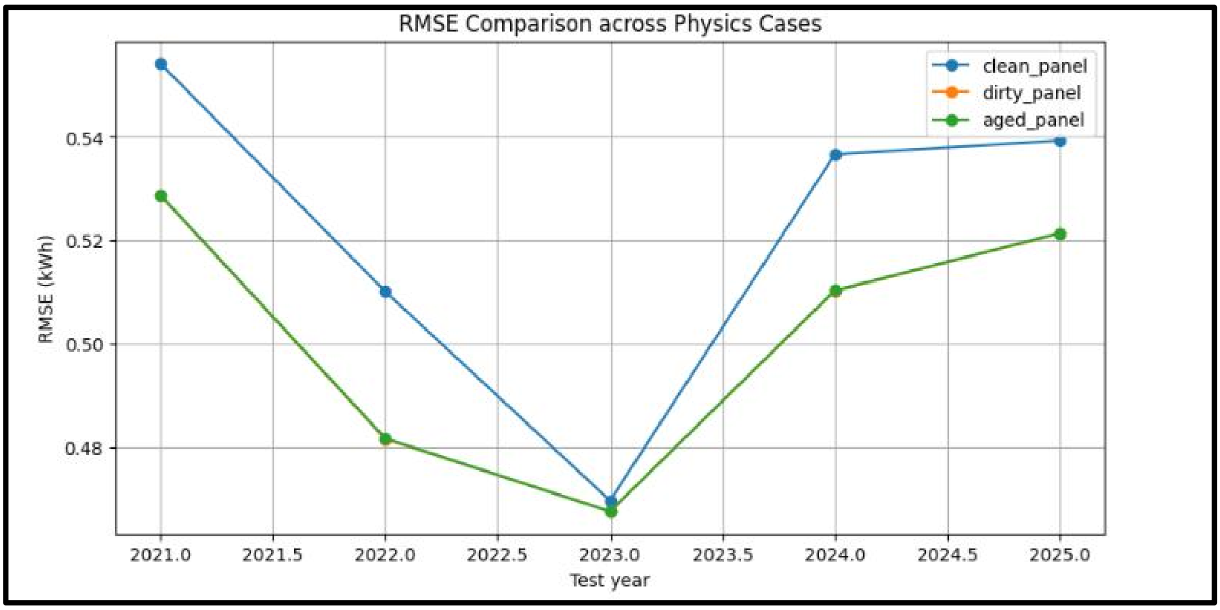

Figure 13. *RMSE trend in the backtest year of three physical scenarios*

Figure 13 shows that the RMSE trends of the three physical scenarios remain close across all backtest years, further confirming that the main source of error comes from weather sampling variability rather than the selected physical loss assumptions.

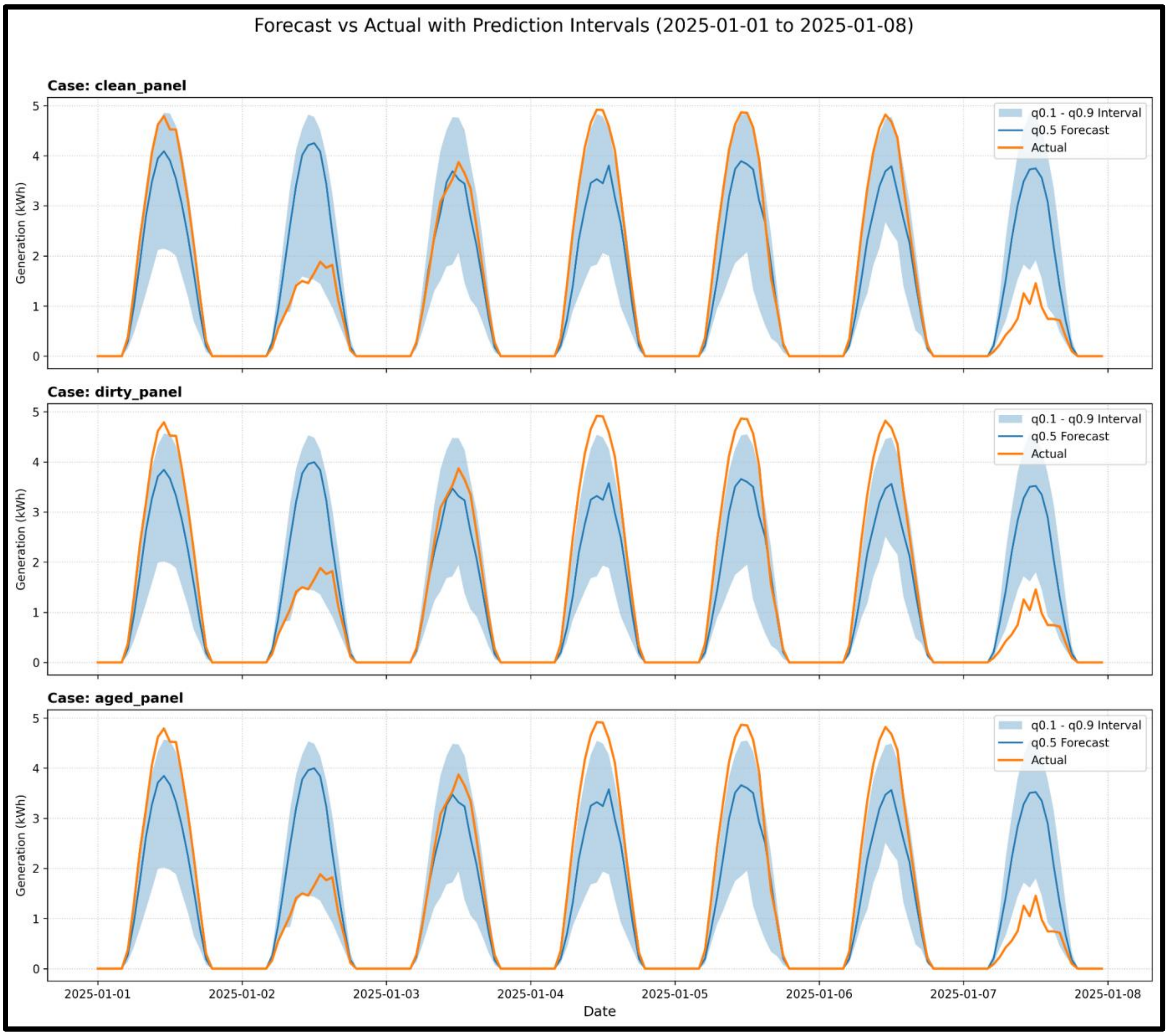

Figure 14. *The time-series graph of three physical scenarios provides a detailed comparison between the actual line, the forecast line (q0.5), and the forecast range (the shaded area between q0.1 and q0.9) over a short period.*

Figure 14 shows the time-series comparison across the three physical cases over the same one-week window highlights the effect of loss parameters on generation magnitude. The dirty panel case consistently produces lower peak generation estimates than the clean panel case, with the aged panel falling between the two. Despite these magnitude differences, all three cases maintain similar interval widths relative to their respective forecasts, indicating that the confidence band scales proportionally with the applied loss factors.

2.5. End-to-end simulation pipeline:

Input and Location Resolution

The end-to-end simulation pipeline receives two user inputs: postcode and installed solar system capacity in kWp. The postcode is mapped to a representative solar reference station using the preprocessed postcode lookup table. This table provides the corresponding *solar_ref_id*, *ref_lat*, and *ref_long*, ensuring that the simulation uses the same representative station mapping as the forecasting pipeline.

Historical Weather Collection and Cleaning

The pipeline automatically determines the forecast period from the current system date. The forecast starts from the current date and covers the next 365 days. Based on this forecast year, the pipeline retrieves historical hourly weather data from previous years using the representative station coordinates. The selected variables include *ALLSKY_SFC_SW_DWN*, *T2M*, *WS2M*, and *RH2M*. Invalid values such as -999 are replaced with missing values and cleaned using interpolation, forward fill, and backward fill.

Scenario Generation

For each date in the 365-day forecast horizon, the system samples multiple historical 24-hour weather blocks with similar day-of-year characteristics. This creates a set of plausible future weather scenarios without requiring actual future weather data or a deterministic weather forecast.

Solar Generation Simulation

After the future weather scenarios are generated, each sampled weather scenario is converted into hourly solar generation using the physics-based solar model. The calculation applies the user-provided system capacity, performance ratio, NOCT-based temperature derating, soiling loss, and degradation loss.

Forecast Output

The simulated solar generation scenarios are aggregated into probabilistic forecasts using q0.1, q0.5, and q0.9. The final output includes an hourly solar generation forecast for the next 365 days, an 80% prediction interval, a monthly aggregated forecast with P10–P90 error bars, and CSV files for downstream energy planning.

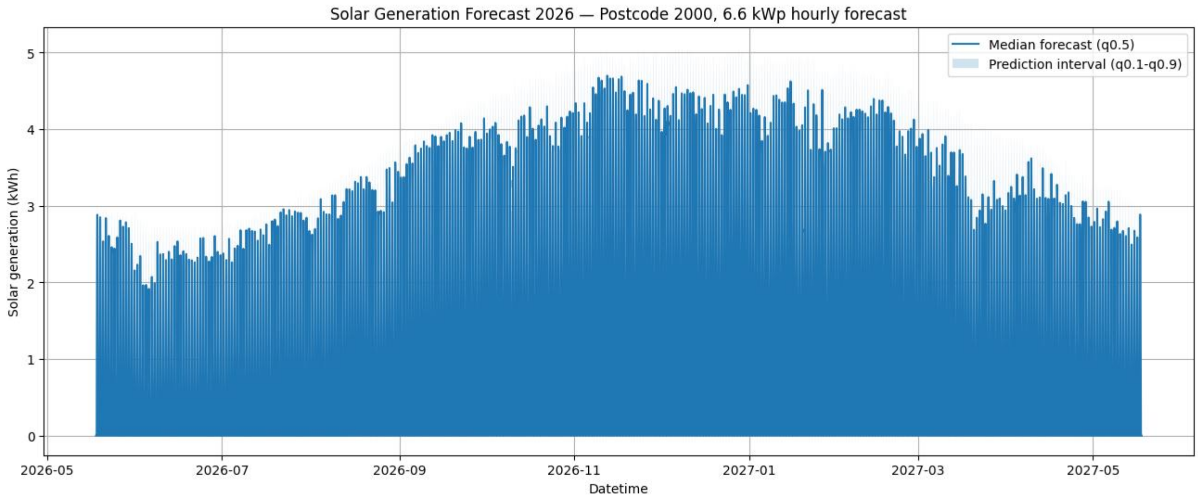

Figure 15*. Full-year hourly solar generation forecast for postcode 2000 with a 6.6 kWp solar PV system. The blue line represents the median forecast (q0.5), while the shaded region shows the prediction interval between q0.1 and q0.9.*

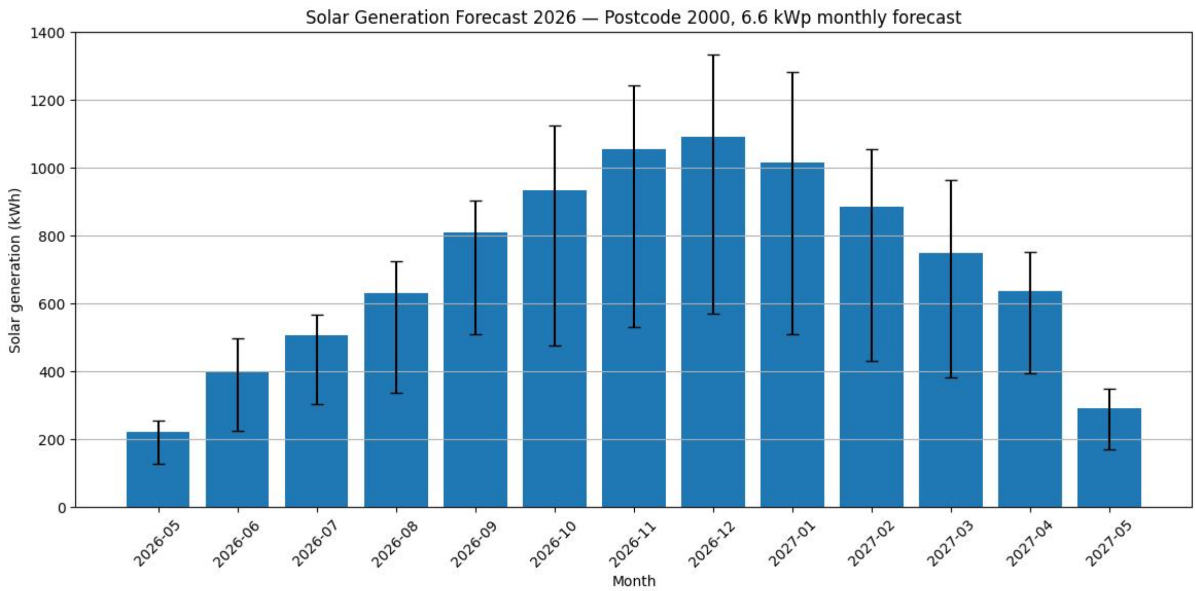

Figure 16. *Monthly aggregated solar generation forecast for postcode 2000 with a 6.6 kWp solar PV system. The bars represent the median monthly forecast (q0.5), and the error bars indicate the uncertainty range from q0.1 to q0.9.*

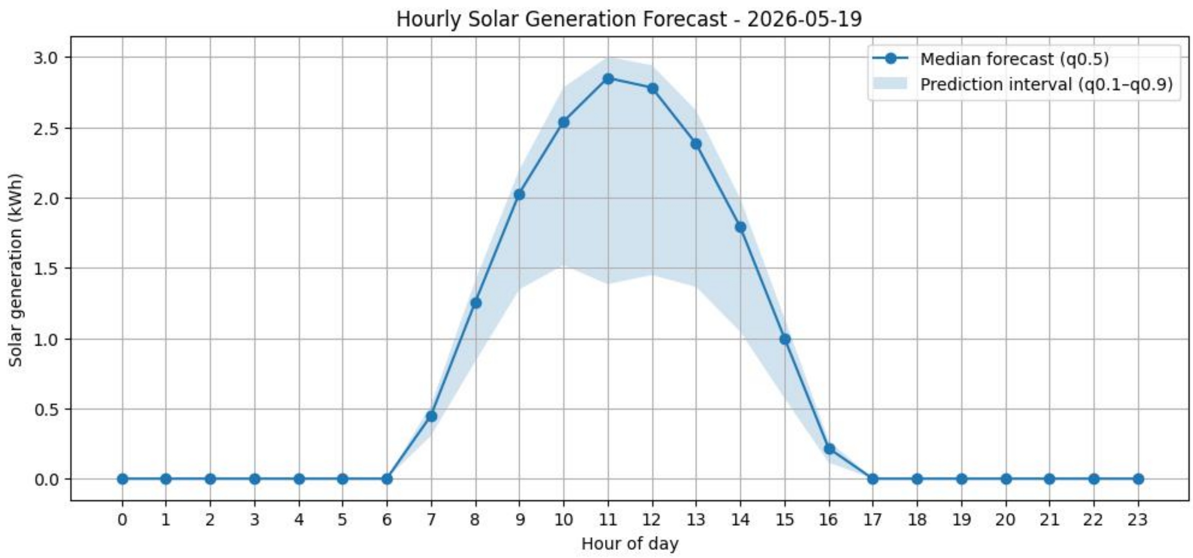

Figure 17. *One-day hourly solar generation forecast for 2026-05-19. The curve shows the median hourly forecast (q0.5), while the shaded region represents the prediction interval between q0.1 and q0.9.*

III. Conclusion

This project developed two complementary approaches for solar power generation forecasting: a fine-tuned Chronos-2 forecasting pipeline and a scenario-based simulation pipeline.

The Chronos-2 approach provides a data-driven forecasting solution that can capture historical patterns and relationships between solar generation, weather variables, and temporal features. With future covariates such as solar radiation, temperature, wind speed, and relative humidity, the fine-tuned Chronos-2 model can produce accurate long-term forecasts. However, this approach depends on the availability or estimation of future weather data, which can be difficult in real deployment scenarios.

In contrast, the scenario-based simulation approach does not require direct future weather inputs. Instead, it generates possible future weather scenarios by sampling historical weather blocks with similar seasonal characteristics, then converts them into solar generation using a physics-based model. Although this approach may have lower point-forecast accuracy than a data-driven model with reliable future weather covariates, it is less constrained by the need for exact future weather data and provides useful probabilistic ranges for uncertainty-aware energy planning.

Overall, the two approaches serve different but complementary purposes. Chronos-2 offers strong forecasting performance when future weather covariates are available or can be estimated reliably, while scenario-based simulation provides a more flexible alternative when future weather data is unavailable. Together, they provide both accurate forecasting capability and practical uncertainty analysis for household energy planning, battery storage optimization, and EV charging decisions.